ALICE L WALTON SCHOOL OF MEDICINE

Glucose-6-Phosphate Dehydrogenase and Pyruvate Kinase Deficiencies

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Adapted from: Glucose-6-Phosphate Dehydrogenase Deficiency

Listen to this Brick

0:00 / 28:00 1x

Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of the mentioned anemia.
2. Explain the pathophysiology of the mentioned anemia.
3. Interpret cli[illegible] mentioned anemia.
4. Outline the management of the mentioned anemia.
5. Distinguish G6PD from pyruvate kinase (PK) deficiency.

CASE CONNECTION

JR is a 24-year-old female who returns to the office 4 days after being diagnosed with an uncomplicated urinary tract infection and prescribed trimethoprim-sulfamethoxazole (TMP/SMX). "I took the medicine exactly like I was told, and I feel worse. First, I turned yellow, and now I'm feeling awfully tired," she reports. On examination she is jaundiced and has icteric sclerae. On palpation during physical exam, the liver is nontender and a normal size. The remainder of the examination is normal.

What is the likely diagnosis? What tests would you order to confirm the diagnosis? Consider your answers as you read, and we'll revisit JR at the end of the brick.

GO TO CONCLUSION ↓

What Is Glucose-6-Phosphate Dehydrogenase Deficiency?

Glucose-6-phosphate dehydrogenase (G6PD) deficiency is a hereditary condition caused by mutations in the G6PD gene that disrupt the proper functioning of red blood cells (also called erythrocytes). Specifically, the red blood cells (RBCs) be[illegible]injury and are easily lysed (hemolyzed).

This is a type of **hemolytic anemia**, which occurs when RBCs are being destroyed faster than new RBCs can be produced. Recall that in general, hemolysis can be due to metabolic changes in the red blood cell (as in this condition), increased RBC fragility, genetic abnormalities in the RBC membrane, immune destruction of RBCs, metabolic changes in the blood including oxidative stress, or physical damage to RBCs as they circulate through a damaged vascular system.

In G6PD deficiency what happens to red blood cells (RBCs)?

Epidemiology. The prevalence of G6PD deficiency varies widely around the world, and it is less than 1% in most regions. It is most common in individuals with genetic roots in the Mediterranean (e.g., Greece, Spain, Italy, Turkey, the Middle East) and in some tropical regions (e.g., Nigeria, Thailand). In some groups (Kurdish Jews[illegible]. Approximately 10% of Black people in the United States are deficient in this enzyme, but this depends on their region of genetic origin.

Note that this condition bears some similarity to sickle cell disease (and trait) in that affected patients are relatively more resistant to malaria than those who are not affected. There is some overlap with the geographic regions of common prevalence as well (e.g., tropical regions).

CLINICAL CORRELATION

There are certain disorders that confer immunity to malaria: G6PD deficiency, sickle cell disease (and trait), pyruvate kinase deficiency, and hemoglobin C. Why would these patients be resistant to this blood-borne intracellular pathogen? It is likely due to an inhospitable environment of red blood cells, so the RBCs lyse before the parasite that causes malaria can reproduce.

How Do Patients with Glucose-6-Phosphate Dehydrogenase Deficiency Present (LO #1)?

Patients with G6PD deficiency are often **asymptomatic** and are not anemic if triggers (oxidizing agents) that cause symptoms are successfully avoided.

Hours after ingesting an oxidizing agent, however, patients may present with symptoms of a **hemolytic crisis**, including dark urine (hemoglobinuria), yellow skin (jaundice), acute kidney injury (rising serum creatinine), and back or flank pain (due to renal injury from hemoglobin in the urine).

Patients may not pres[illegible]xposed to triggers.

What are typical symptoms during hemolysis due to G6PD deficiency?

What Is the Pathophysiology of Glucose-6-Phosphate Dehydrogenase Deficiency (LO #2)?

G6PD deficiency is an inherited disorder, but its impact on a patient's life can vary depending on the degree of expression and the way the disease is managed. Before we go into the details of G6PD deficiency, let's review the role of G6PD in protecting RBCs from molecules that could be harmful.

Glutathione Pathway and Oxidative Damage

Free radicals are steadily produced in the red blood cells. A free radical is a molecule that has an unpaired electron that is ready to react with other substances. When th[illegible]ee radical reacts with is said to oxidize, often causing damage to the body (including RBCs) in the process.

If this oxidation were left unchecked, all of our RBCs would experience so much **oxidative damage** that they would ultimately undergo rapid lysis (hemolysis). So what do our RBCs have to counteract this oxidation? The antioxidant molecule **glutathione**.

Glutathione has two states: the reduced state (2 GSH) and the oxidized state (GS-SG) (Figure 1).

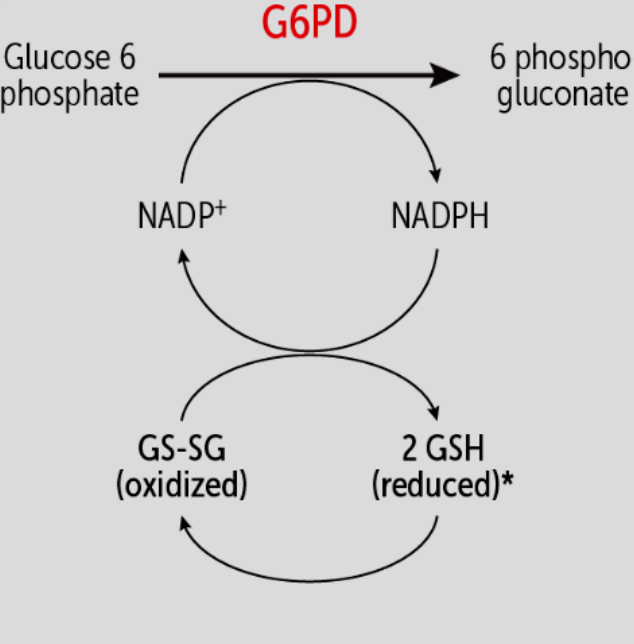

Figure 1

The reduced state of glutathione protects red blood cells from oxidative damage from hydrogen peroxide, by detoxifying the hydrogen peroxide, turning it into water. Therefore, it is imperative to maintain an adequate amount of reduced glutathione in our RBCs. To keep glutathione in its reduced state, however, the RBCs need the molecule NADPH, the reduced state of $NADP^+$. This is where the enzyme G6PD comes in: G6PD keeps NADPH (and glutathione) in their reduced states. This protects RBCs from oxidative damage.

Why is G6PD necessary to prevent free radical–induced hemolysis?

Etiology and Pathophysiology

In G6PD deficiency there is a complete or partial absence of G6PD. The resultant lack of NADPH decreases reduced glutathione, and ultimately there is an increase in free ra[illegible]Cs have enough of the enzyme to manage the free radical buildup that occurs during normal cell metabolism. But during times of oxidative stress there is not enough G6PD to counteract all the free radicals, and cellular damage occurs. In addition, the level of G6PD activity decreases over time. This means that older RBCs are more susceptible to damage and their lifespan is shorter. What are some of these stressors? Oxidizing agents include:

- Fava beans (common in Mediterranean countries)
- Drugs such as sulfonamides, dapsone, cotrimoxazole, aspirin, primaquine, nitrofurantoin, chloramphenicol, and rasburicase
- Infections (particularly parvovirus B19) can also have an oxidizing effect, both causing oxidation and hemolysis and impairing the bone marrow's ability to produce RBCs to compensate for those that are destroyed.
- Oxidative damage to the membrane of red blood cells can be so severe that the RBCs burst open within blood vessels (hemolysis). The hemolysis that occurs in G6PD deficiency is mainly **intravascular hemolysis** (within blood vessels).

Genetics

G6PD is an inherited disorder that is passed down the generations in an **X-linked pattern**. This means that males (with a single, affected X chromosome) usually manifest the abnormality, while females are carriers. Female carriers are almost always heterozygous (one X chromosome that carries the mutation and one that doesn't) and are unlikely to show symptoms. Females may be symptomatic if they are homozygous or if inactivation of their normal X chromosome occurs.

Penetrance of this dis[illegible] a partial G6PD deficiency may have [illegible]th more complete G6PD deficiency may present with more prominent symptoms.

Pathology

What morphologic changes are present in the blood of a patient with G6PD deficiency? We can visualize RBC inclusions called **Heinz bodies** with vital stains (Figure 2).

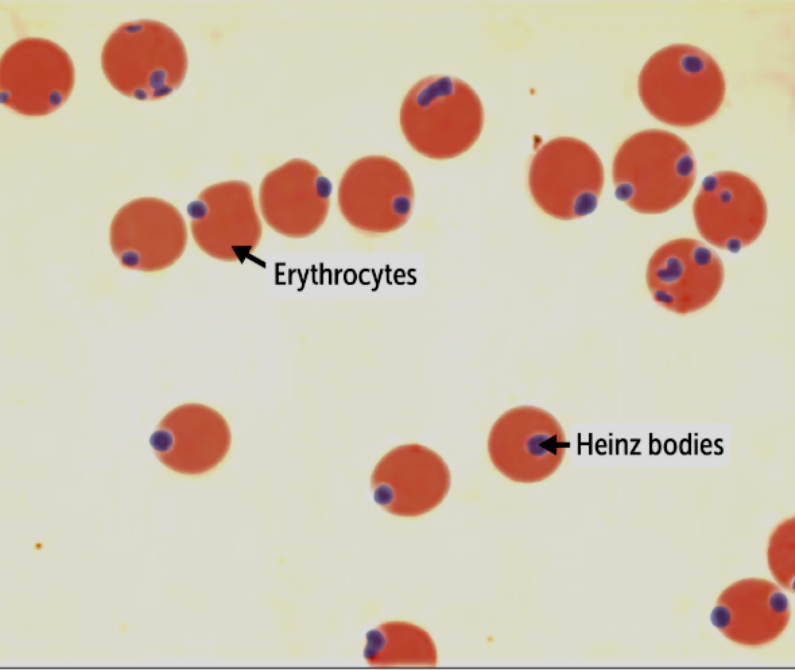

Figure 2

Heinz bodies occur w[illegible]s and cause the dissociation of hemoglobin into heme and globin. Globin in its denatured form then separates into a clump that attaches to the inside of the red blood cell membrane. Spherocytes, bite cells, and RBC fragments may also be seen on a Wright-Giemsa–stained peripheral blood smear at the time of a hemolytic episode.

As these red blood cells with Heinz bodies pass through the spleen, macrophages recognize them as abnormal; they may not destroy the RBCs entirely, but rather engulf and remove the membrane to which the Heinz body is attached. This membrane damage causes the formation of the characteristic **bite cells** (Figure 3) of G6PD deficiency.

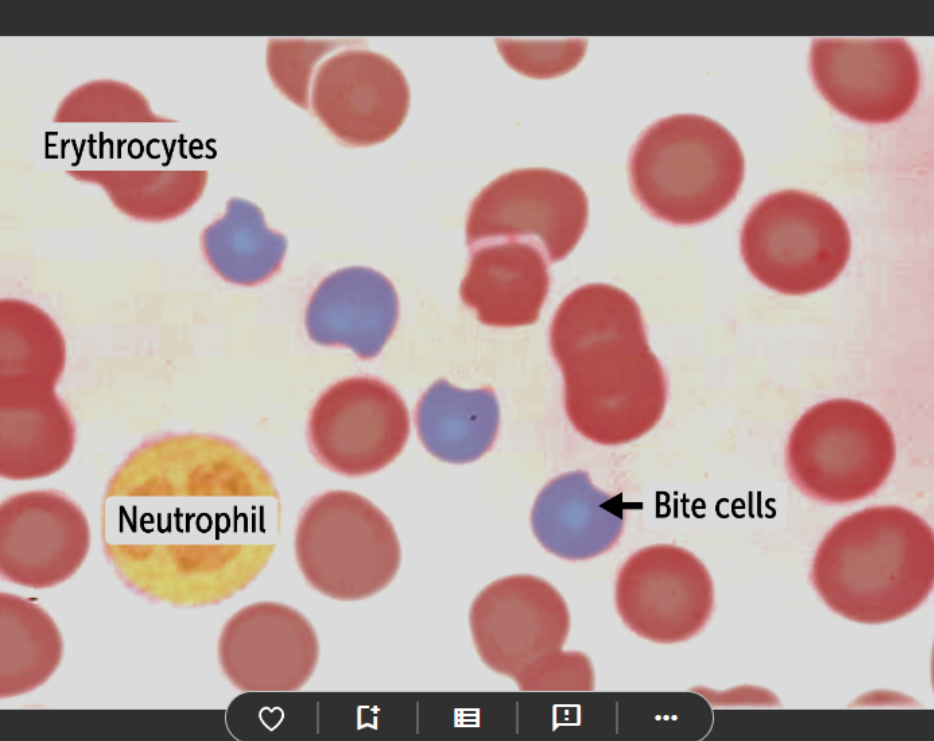

Figure 3

As the bite cells continue to circulate through the body their shape can become less deformed and more rounded, producing **spherocytes**. These have reduced membrane surface and therefore less flexibility, so spherocytes are often destroyed in the spleen by **extravascular hemolysis.**

CLINICAL CORRELATION

Spherocytes are also seen in autoimmune hemolytic anemia and hereditary spherocytosis.

What are morphologic changes present in the blood of a patient with G6PD deficiency?

How Do We Diagnose Glucose-6-Phosphate Dehydrogenase Deficiency (LO #3)?

G6PD deficiency may be diagnosed at any age, but it frequently becomes symptomatic in childhood. Investigations usually begin following an acute episode, sometimes triggered by an identifiable event or exposure.

Physical Examination

Patients may have a normal exam, or, if in a crisis, may present with signs of hemolysis such as jaundice, hemoglobinuria, and back or flank pain. They may also have general signs of anemia (e.g., fatigue, tachycardia, dyspnea).

Laboratory Tests

In the diagnosis of G6PD deficiency, laboratory tests are conducted both during and after a hemolytic crisis. Findings during the crisis are similar to those in other types of hemolytic anemia. The confirmatory diagnosis comes with special studies done after a crisis is over.

Studies done during the hemolytic crisis. During an acute hemolytic episode, findings from a complete blood count (CBC) include a low hemoglobin (Hb) and hematocrit (Hct) with a normal mean corpuscular volume (MCV). These are generally normal when a crisis is not occurring.

As in other hemolytic anemias, a liver function panel will show evidence of hemolysis. This means increased indirect (unconjugated) bilirubin, low haptoglobin, and elevated lactate dehydrogenase (LDH).

During a hemolytic crisis, the urinalysis may show heme (due to the filtered hemoglobin). The serum creatinine may be elevated due to acute kidney injury from the toxic effect of the filtered hemoglobin on the renal tubular cells.

In G6PD deficiency what are the laboratory findings during an acute hemolytic episode?

Confirmatory studies after the hemolytic crisis. The diagnosis of G6PD deficiency is confirmed with enzymatic studies that measure the level of G6PD activity present in the red blood cells. The levels will be low in patients with G6PD deficiency. The study should **not** be done during or immediately after a hemolytic crisis, since the newly made RBCs and reticulocytes have a relatively higher G6PD activity, so will mask the reduced enzyme activity characteristic of the disorder. It is recommended to wait at least 3 months after a hemolytic episode to evaluate for G6PD deficiency (the life cycle of RBCs is 120 days; this timeframe allows for the blood to have a new batch of red cells).

When is the proper time to obtain enzymatic studies in a patient with G6PD deficiency?

How Do We Manage Glucose-6-Phosphate Dehydrogenase Deficiency (LO #4)?

The easiest, most practical treatment for this disease is **avoiding triggers**, such as oxidizing agents (e.g., sulfonamides, antimalarial drugs, hydrogen peroxide, aspirin, fava beans). Of course, not all infections can be avoided, so patients should be coached to be aware of signs of infection.

Most hemolytic events are self-limiting and do not lead to severe anemia. This is because the newer red blood cells can be made rapidly, and they have relatively more G6PD than the older RBCs. Treatment is unnecessary in these cases.

However, if the hemoglobin levels are dangerously low, **RBC transfusion** is appropriate. Patients[illegible]n oral or IV volume repletion to prevent the [illegible] effects of hemoglobinuria.

What triggers should patients with G6PD deficiency avoid?

Pyruvate Kinase Deficiency (LO #5)

Pyruvate kinase (PK) deficiency is a rare, autosomal recessive condition, occurring most commonly in patients of Northern European descent.

As mentioned in previous bricks, pyruvate kinase catalyzes the final step in glycolysis: the irreversible transfer of phosphate to ADP, generating ATP. This lack of ATP makes red cells more rigid and more likely to hemolyze. Hemolysis in PK deficiency is mainly extravascular, where red cells are phagocytized (eaten) by macrophages in the spleen. Additionally, PK deficiency causes an increased level of 2,3-BPG, leading to a decreased affinity of hemoglobin for oxygen.

The clinical presentation [illegible] is highly variable, ranging from a mild, fully compensated chronic hemolysis without anemia to life-threatening transfusion-dependent hemolysis present at birth. Patients with severe hemolysis may develop complications of chronic hemolytic stages, such as extramedullary hematopoiesis, gallstones, and iron overload.

CLINICAL CORRELATION

Pyruvate kinase (PK) deficiency is an important diagnosis to keep in mind when hemolytic anemia presents in a newborn patient.

There are no characteristic RBC morphologic findings in PK deficiency; however, nonspecific changes such as echinocytes (burr cells) may be present. The diagnosis is typically made by either enzyme assay or genetic testing.

Treatment strategies for this deficiency vary widely, from no therapy to chronic red cell transfusions. Due to the risk of iron overload in patients with chronic transfusions, iron status is monitored regularly with addition of iron chelation therapy if necessary. Splenectomy has documented benefits in severe cases.

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to JR, what is the clinical diagnosis? What tests were ordered to confirm the diagnosis?

A complete blood co[illegible] hemoglobin. On peripheral smear, bit[illegible]en. Additionally, the total bilirubin is elevated, and it is mostly indirect. From the clinical presentation and this blood work, acute hemolysis due to glucose-6-phosphate dehydrogenase (G6PD) deficiency triggered by the use of the antibiotic TMP/SMX is suspected. This diagnosis is explained to JR and she is instructed to immediately stop the drug. JR follows up in 2 weeks; she is doing well. At this visit, G6PD enzyme activity level is checked and it is low. You discuss with JR important triggers to avoid and urge her in the future to mention this issue to any physician when prescription medications are ordered.

Summary

- Glucose-6-phosphate dehydrogenase (G6PD) deficiency is a hereditary condition that disrupts the functioning of red blood cells (RBCs), making them more susceptible to oxidative injury and hemolysis.
- G6PD deficiency is more prevalent in individuals with genetic roots in the Mediterranean and some tropical regions.
- Patients with G6PD deficiency may be asymptomatic, but can present with symptoms of a hemolytic crisis, including dark urine, yellow skin, and back or flank pain.
- G6PD is necessary to generate NADPH, which is crucial for the production of reduced glutathione, an antioxidant molecule that protects RBCs from oxidative damage.
- Definitive diagnosis of G6PD deficiency requires enzymatic studies that measure the levels of G6PD activity in circulating RBCs.
- Management of G6PD deficiency primarily involves avoiding triggers, such as oxidizing agen[illegible]self-limited and do not require treatm[illegible]cessary for patients with dangerously low hemoglobin levels.
- Pyruvate kinase (PK) deficiency is a rare, autosomal recessive condition in patients of Northern European descent in which lack of ATP causes rigid RBCs and chronic extravascular hemolysis. Clinical symptoms range from mild chronic hemolysis to life-threatening, transfusion-dependent hemolysis. Diagnosis is made with enzyme assay or genetic testing. Management of PK deficiency is variable, from no therapy to chronic transfusions.

Review Questions

1. A 24-year-old male has experienced lower back pain and dark urine for the past day after a meal of fava beans. He recalls that his grandfather avoided eating them because they make him "feel ill." His uncle had a similar painful episode when he was a child after he was treated for a urinary tract infection. The patient's eyes appear yellow. The complete blood count (CBC) shows anemia but normal leukocytes and platelets. Which of the following types of anemia is most likely in this patient?

A. ○ Aplastic anemia
B. ○ Macrocytic anemia
C. ○ Megaloblastic anemia
D. ○ Microcytic hypochromic anemia
E. ● Normocyt[illegible]

Hide Explanation

The correct answer is normocytic anemia (E). Given the patient's anemia and evidence of hemolysis (icterus, back pain) following an oxidizing agent ingestion (fava beans), and the family history of a similar episode, the patient likely has glucose-6-phosphate dehydrogenase (G6PD) deficiency, a form of hemolytic anemia. Hemolytic anemias generally have a normal mean corpuscular volume (MCV) (normocytic), but with a high variety of red blood cell sizes (high RBC distribution width [RDW]). Aplastic anemia (A) is a decrease in all blood cell lineages due to decreased production from the bone marrow. The normal white blood cell (leukocyte) and platelet count here makes this unlikely. Macrocytic anemia (B) is incorrect because the mean corpuscular volume of RBCs is not increased in hemolytic anemias, including G6PD deficiency. Megaloblastic anemia (C) is caused by vitamin B12 and folate deficiency, which is not consistent with this history. Microcytic hypochromic anemia (D) is seen in iron deficiency anemia, which is not consistent with this history.

2. A 7-year-old, healthy male is brought to the pediatrician after genetic analysis reveals that the child has G6PD deficiency. Which of the following is the best treatment for this male?

A. ○ Bone marrow transplant
B. ○ Corticosteroid drugs
C. ● Trigger a[illegible]
D. ○ Vitamin E
E. ○ Weekly blood transfusions

Hide Explanation

The correct answer is trigger avoidance (C). The male is otherwise healthy, and he will not experience signs or symptoms of the disease unless he ingests disease-triggering substances, such as fava beans and certain drugs. So the best treatment for this patient is to simply avoid these substances. Bone marrow transplant (A) is used as a last resort for some very severe types of anemia, but it is unnecessary in glucose-6-phosphate dehydrogenase (G6PD) deficiency, as trigger avoidance prevents hemolytic episodes. Corticosteroid drugs (B) are used in some types of anemia, particularly autoimmune hemolytic anemia, but are not effective in G6PD deficiency. Vitamin E (D) is not involved in the underlying mechanism of G6PD deficiency and does not alleviate the symptoms. Weekly blood transfusions (E) are unnecessary because the patient will not be anemic unless he is exposed to an oxidizing agent.

3. Which of the following substances is most likely decreased in a patient with glucose-6-phosphate dehydrogenase (G6PD) deficiency?

A. ○ GS-SG
B. ○ Glucokinase
C. ○ Glucose-6 phosphatase
D. ● Glutathione
E. ○ Pyruvate kinase

Hide Explanation

The correct answer is glutathione (D). Glucose-6-phosphate dehydrogenase (G6PD) deficiency causes an inability to replenish reduced glutathione, reducing its level and rendering cells susceptible to damage by free radicals. Oxidized glutathione (GS-SG [A]) therefore increases in the cells. Glucokinase (B) is an enzyme of glycolysis. Glucose-6-phosphatase (C) deficiency causes von Gierke disease, a glycogen storage disease. Pyruvate kinase (E) deficiency is the incorrect answer as pyruvate kinase is not affected in patients with G6PD deficiency.

4. A 1-month old male presents to the hospital with fatigue, irritability, elevated bilirubin levels and clinical jaundice. His CBC demonstrates a marked normocytic anemia (hemoglobin 6.0 g/dL, normal: 10.0-17.0 g/dL) and a reticulocyte count is increased. The inpatient team suspects a diagnosis of pyruvate kinase deficiency. Which of the following treatments would be beneficial?

A. ○ Iron supplementation
B. ○ Corticosteroid drugs
C. ○ No treatm[illegible]
D. ● Red blood cell transfusion
E. ○ Splenectomy

Hide Explanation

The correct answer is red blood cell transfusion (D). In this patient with pyruvate kinase deficiency, correcting the severe anemia with red cell transfusion is most important in this initial phase. It is likely the patient will require chronic transfusions for his treatment. Iron supplementation (A) is not required as the patient does not have iron deficiency. Corticosteroid drugs (B) are not a documented treatment for pyruvate kinase deficiency. No treatment (C) would not be beneficial in this case, as it does not address his severe symptomatic anemia. Splenectomy (E) may be helpful in severe cases. However, it is not recommended until after the age of 3, as the degree of hemolysis may decline after infancy and the risk of infection with certain organisms declines.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of the mentioned anemia.~~
02 ☑ ~~Explain the pathophysiology of the mentioned anemia.~~
03 ☑ ~~Interpret clinical findings to properly diagnose the mentioned anemia.~~
04 ☑ ~~Outline the management of the mentioned anemia.~~
05 ☑ ~~Distinguish G6PD from pyruvate kinase (PK) deficiency.~~

Objective Confidence: 5/5